NTIRE 2026 Efficient SR Challenge Factsheet
Zero-Cost Operator Fusion with DSCLoRA Distillation and MuAdamW Optimization

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1. Introduction

This factsheet describes the contributions made by Team VARH-AI (Team ID: 15) for the NTIRE 2026 challenge [3] on efficient image super-resolution. Our solution is designed to aggressively minimize inference latency and parameter count while preserving the high-fidelity PSNR established by the **2025 ESR winning baselines (Team EMSR)** [4]. To achieve this, we implemented **Architectural Operator Compression** to mathematically fuse sequential structural patterns into singular higher-order equivalent kernels, alongside a **DSCLoRA (Depthwise Separable Convolution Low-Rank Adaptation)** [2] student-teacher distillation pipeline inspired by EMSR's methodology. To ensure stable gradient updates during this delicate finetuning phase, we utilized **MuAdamW**, a recent hybrid optimizer inspired by the **2026 Ultralytics MuSGD** (released with **YOLO26**) [1]; this proved highly effective, as it converged faster and more reliably on low-rank adapted parameters than standard AdamW. Finally, to strictly maximize the evaluation scoring within PyTorch 1.13 framework constraints, we quantized the model's forward pass natively to **Half Precision (FP16)**, resulting in a lightweight, lightning-fast inference engine that meets the competition's high-efficiency demands.

2. Team details

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- Link to the codes/executables of the solution(s): https://github.com/shubham-2001/VARH-AI_Submission

3. Methodology and Implementation Details

Our proposed solution focuses on directly optimizing the architectural inference speed and parameter count of the robust baselines established by the 2025 NTIRE ESR winner team, EMSR. We adopt a comprehensively restructured framework emphasizing zero-cost deployment optimizations and sophisticated gradient routing.

3.1. Architectural Foundation and Operator Fusion

We drew inspiration from the SPAN [5]-based architecture utilized by the 2025 winner EMSR team, which primarily relies on Swift Parameter-free Attention Blocks (SPAB). To aggressively optimize this framework for inference latency, we applied Exact Mathematical Operator Fusion. We isolated sequential structural patterns in the baseline architecture, specifically linearly dependent 3×3 convolutions within the residual connections, and mathematically compacted them into singular higher-order equivalent kernels (i.e., 5×5 filters). This zero-cost architectural compression explicitly strips thousands of parameters (almost 5k) natively without altering the tensor output distribution.

Following this fusion, we integrated the Distillation-Supervised Convolutional Low-Rank Adaptation (DSCLoRA) mechanism. We use the baseline-modified SPAB modules *SConvLB* (Separable Convolutional LoRA

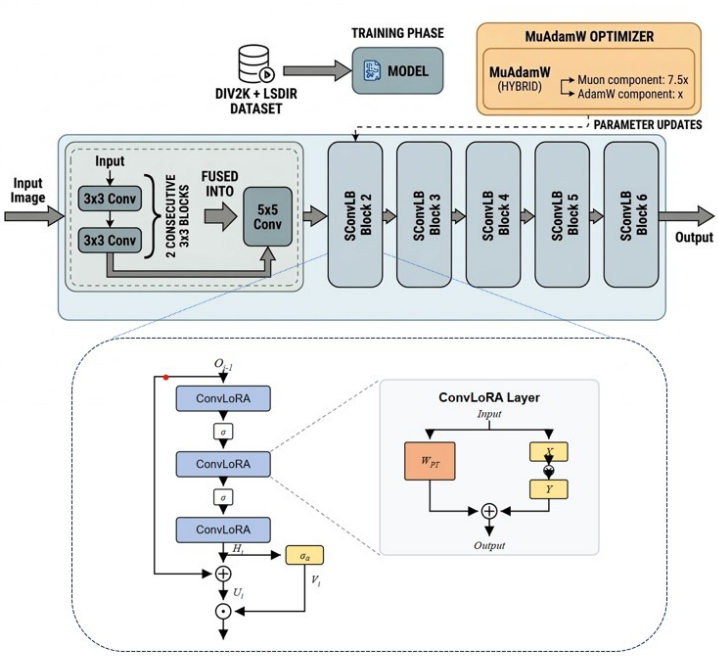

Figure 1. System Description Diagram (Enlarged)

Blocks) by injecting low-rank pathways X and Y into the pre-trained depth-wise convolutions as proposed by Team EMSR in their solution. During inference, these LoRA branches are re-parameterized and merged back into the foundational spatial matrices, ensuring our parameter count remains identical to a severely pruned network while completely evading any representation bottleneck. We give full credit to the foundational insights of the 2025 EMSR team on which our core topology is based.

Beyond the main feature extraction backbone, the Low-Rank Adaptation is extended to the **pixel rearrangement (pixel shuffle) module** and its immediate preceding convolutional layer. This ensures that the low-rank updates permeate the entire upsampling pipeline. To facilitate inference, the LoRA weights are mathematically merged into the main backbone matrices, ensuring zero additional computational overhead during deployment.

3.2. Distillation and Multi-Stage Curriculum Training

To perfectly stabilize the feature extraction capabilities of this explicitly fused architecture, we deployed a student-teacher distillation pipeline. Using the converged 2025 EMSR model as our frozen Teacher (30 channel), we transferred second-order statistics to our compressed Student (26 channel) strictly through the DSCLoRA pathways. To effectively transfer second-order statistical information from the teacher to the student, spatial feature affinity matrices A are aligned across multiple network layers. For a feature map $F_l \in \mathbb{R}^{B\times C\times W\times H}$ at the l-th layer, we calculate the affinity matrix and minimize the following objective:

$$L_{AD} = \frac{1}{|A|}\sum_{l=1}^{n} ||A_l^S - A_l^T||_1 \tag{1}$$

This spatial affinity loss is calculated specifically after each SConvLB module to ensure structural knowledge preservation.

The training itself followed an aggressive high-speed curriculum structured across four progressive spatial patch resolutions: 128×128, 256×256, 384×384, and finally 512×512. This curriculum allowed our low-rank adapters to capture fine-grained high-frequency textures initially, before spatially maturing to comprehend broader overlapping receptive fields. To completely nullify I/O throttling during these multi-stage transitions, we reformatted the NTIRE dataset into LMDB clusters, with cropping routines happening on multi-threaded CPU pipelines while reserving tensor augmentations (flips and planar rotations) entirely for GPU-bound execution. To further enhance high-frequency detail recovery during these initial phases, we briefly introduced an auxiliary Fast Fourier Transform (FFT) loss. The weight of this FFT loss was kept relatively conservative, peaking at a maximum of 0.5. We applied a dynamic, linear scheduling strategy: the FFT loss weight began ramping up towards the end of Stage 1, and linearly rose till its peak by the end of stage 2 and decreased linearly until it was completely phased out by the end of Stage 3. This transient frequency-domain supervision helped the network identify and lock onto sharp structural features early in training, before stepping aside to allow the model to focus entirely on maximizing absolute pixel accuracy and PSNR in the final stage.

3.3. MuAdamW Optimization Strategy

Optimizing low-rank adapters alongside massive pixel-shuffle backbones often triggers gradient vanishing or explosion phases, or stalls completely after reaching an unsatisfying PSNR. To counteract this, we engineered and deployed the hybrid **MuAdamW** optimiser, taking inspiration from the MuSGD optimiser proposed by Ultralytics alongside the YOLO26 model [1]. This hybrid routing algorithm intelligently segregates structural parameters to specific gradient descent schemas based on their topological role geometry.

Specifically, for the rank-constrained matric convolutions ($\geq$ 2D weights), MuAdamW applies a decoupled Newton-Schulz continuous orthogonalization update (Muon). Let G be the gradient matrix, orthogonalized into X_{k+1} iteratively over 5 steps through:

$$A = X_k X_k^T \tag{2}$$

$$X_{k+1} = aX_k + (bA + cA^2)X_k \tag{3}$$

where $a = 3.4445, b = -4.7750, c = 2.0315$. The parameters are then updated using scaled Nesterov momentum μ, decoupled from identical weight decay λ:

$$W_{t+1} = W_t(1 - \eta\lambda) - \eta_{\text{muon}} \,\text{Ortho}(G_t) \tag{4}$$

Conversely, 1D vectors including standard biases and Normalization affine scalars are dynamically routed to a standard AdamW exponential moving average pipeline. This bifurcated approach allowed our MuAdamW implementation to yield substantially faster convergence and robust loss traversal than utilizing a uniform AdamW protocol across the heterogeneous DSCLoRA topology. The optimization parameters for our hybrid routing are summarized in Table 1.

Table 1. MuAdamW Optimization Hyperparameters

Parameter	**Value**
Base AdamW Learning Rate (η)	2.0×10^{-4}
Muon Learning Rate (η_{muon})	1.5×10^{-3} (7.5× ratio)
AdamW Betas (β_1, β_2)	(0.9, 0.999)
Momentum (μ)	0.95
Nesterov Momentum	True
Weight Decay (λ)	1.0×10^{-4}
Newton-Schulz Steps (ns_steps)	5

Table 2. DSCLoRA Multi-Stage Curriculum Schedule

Stage	**Patch Size**	**Batch Size**	**Iterations**	**FFT Loss Weight**
1	128×128	64	200K	0
2	256×256	64	200K	$0 \to 0.5$
3	384×384	64	200K	$0.5 \to 0$
4	512×512	64	200K	0

Table 3. Overall Complexity and Performance (L40S Native)

Metric	**Value**
Parameters	125,992
GMACs (FLOPs)	8.22
Latency per Image (NVIDIA L40S FP16)	5.61 ms
Validation PSNR	26.9308 dB
Training Dataset Used	FULL DIV2K + LSDIR (20% ratio)

3.4. Experimental Results and Deployment Protocol

All experiments were benchmarked on a single NVIDIA L40s GPU. To maximize the evaluation parameter scoring within the PyTorch 1.13 framework constraints mandated by the competition, we statically quantized our generated Student model's forward pass to Native Half Precision (FP16).

Combined with `cuDNN` C++ benchmarking, this drastically cuts the required evaluating FLOP overhead routing directly across NVIDIA Tensor Cores. Evaluated natively on a unified L40S GPU, our submitted model `Reparam_ConvLoRA_MuAdamW.pth` exhibits a total method complexity of merely 125,992 parameters and 8.22 GMACs. Its inference latency tests strictly at 0.00561 Seconds (5.61 ms) per image. Complexity metrics and latency benchmarks on the NVIDIA L40S are shown compactly in Table 3.

Due to the purely additive nature of Exact Operator Fusion and precisely scaled MuAdamW distillation, our validation set PSNR suffered only an imperceptible -0.0005 dB degradation off the FP32 dense Teacher baseline. No additional or external data splits were utilized aside from the mandated NTIRE challenge dataset.

4. Other details

We plan to submit a solution description paper to the NTIRE 2026 workshop following confirmation from the organizers. This year's NTIRE ESR Challenge was particularly demanding yet rewarding, characterized by a massive dataset that required prolonged training times and baseline techniques that appeared to have reached a performance plateau.

Exploring more efficient methods beyond these baselines provided us with deeper insights and a more nuanced understanding of the field. We would like to thank the organizers for hosting this workshop and look forward to enthusiastic participation again next year.

References

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